

0- Overview of all scripts

- I - Behavioral data processing : data processing for demographics, classification of the patient malformation, cognitive scores (processing speed, working memory) and SRS2
- II - Choosing the control participants : matching with the DCC patients to choose the control participants
- III - SRS-2 Manual Entry & Editing Tool : create a csv that contains answers to SRS2 questions when missing from the RedCap csv

I- Behavioral data processing

Overview

This project implements a pipeline to extract, clean, and analyze behavioral data from REDCap and other sources. It includes multiple behavioral domains:

- Demographics
- Neuroclassification
- Cognitive scores (Processing Speed, Working Memory)
- SRS-2 (Social Responsiveness Scale, Second Edition)

All data are harmonized into a common format to enable downstream statistical analyses and integration with neuroimaging data. All datasets share a common subject identifier (**subj_id**) and can be merged for multimodal analyses.

Project Structure

- RedCap/ # contains all necessary scripts
- RedCap/data/ # Raw REDCap and external datasets
- RedCap/csv/ # Processed outputs

Dependencies

- All scripts run after installing the conda environment from file : XXX TODO
- If you don't want to use conda, make sure to have all these libraries available:
 - Python 3.x
 - pandas
 - numpy
 - openpyxl
 - LibreOffice (headless mode)

How to Run

Main entry point:

```
python extract_behavioral_data.py
```

This script:

1. Extracts all behavioral domains
2. Structures them into clean dataframes
3. Prepares them for merging and analysis

Behavioral Data Extraction (extract_csv.py)

All behavioral data (except SRS-2 scoring) are extracted using functions defined in:

```
extract_csv.py
```

1. Demographics

- `extract_demographics()`

Output:

- subj_id
 - sex
 - age
-

2. fMRI Metadata

- extract_fmri_data()
- Reads: data/fmri.csv
- Keeps only valid acquisitions
- Selects latest scan per subject
- Extracts:
 - acquisition age
 - gender

Output:

- subj_id
 - acquisition_age
 - gender
-

3. Neuroclassification

- extract_neuro_classification()
- Reads: data/Neuro.csv
- Parses CCD column into two levels:

Main classification:

- Total
- Hipoplásico
- Hiperplásico
- Parcial
- Controle

Secondary classification:

- Isolado
- Plus (associated malformations)

Output:

- subj_id
 - neuro_classification
 - isolado_classification
-

4. Cognitive Scores

Cognitive data are extracted from REDCap exports with heterogeneous formats. The pipeline dynamically identifies which test was used and retrieves the correct scores.

a - Generic Extraction Logic

`extract_redcap_general(...)` This function:

- Parses "Instruments used"
- Identifies which test was performed
- Maps each test to the correct column index
- Extracts:
 - Raw scores
 - Scaled scores
- Cleans redundant columns

This function is reused for multiple cognitive domains.

b - Processing Speed

`extract_processing_speed()`

- Reads: `data/ProcessingSpeed.csv`
- Handles multiple instruments:
 - Coding (WISC-IV / WAIS-III)
 - Symbol Search

Outputs include:

- `Coding_Raw`
 - `Coding_Scaled`
 - `Search_Raw`
 - `Search_Scaled`
-

c - Working Memory

`extract_working_memory()`

- Reads: `data/WorkingMemory.csv`
- Extracts:
 - Digit Span
 - Corsi Blocks
- Handles subcomponents:
 - Forward
 - Backward

Outputs include:

- Raw scores
 - Span scores
 - Total scores
 - Scaled scores
-

5 - SRS-2

This project implements a full pipeline to **process, clean, and analyze SRS-2 (Social Responsiveness Scale, Second Edition) data** from two sources:

- **IDOR dataset (patients)**
- **HCP dataset (controls)**

The pipeline extracts raw questionnaire data, harmonizes formats, computes SRS-2 scores using an official scoring sheet, and performs basic statistical analyses (to be continued).

- Supports multiple test versions:
 - Preschool (`pre`)
 - School-age (`escolar`)
 - Adult (`adulto a`, `adulto h`)

- Harmonizes REDCap structure into unified question columns
- Handles missing responses
- Uses official scoring via Excel

Main Features

- Extraction of SRS-2 data from **REDCap exports**
 - Handling of the truplication issue by generating a more easy-to-read format
 - Integration of:
 - Demographic data
 - Neuroclassification labels
 - Handling of missing or invalid responses
 - Automatic computation of:
 - Total score
 - Subscale scores
 - Comparison between **patients and controls**
-

Input Data : required files (in data/)

- `SRS2.csv` : REDCap export (IDOR dataset)
- `SRS2_correction.xlsx` : official scoring sheet
- `srs02_HCP.txt` : HCP dataset (tab-separated)

Data Processing Pipeline

Usage Run the full pipeline:

```
python extract_SRS2.py
```

This will:

1. Extract and format IDOR and HCP datasets
2. Compute SRS-2 scores
3. Merge results from the 2 populations
4. Run basic analyses

1. Extraction and Formatting IDOR dataset (`extract_SRS2`)

- Merges SRS-2 responses with demographic data
- Fixes column inconsistencies (e.g., shifted indices in escolar version)
- Maps REDCap format to unified structure:
 - One column per question (1 to 65/66)
- Keeps only the **latest test per subject**

HCP dataset (`extract_srs2_HCP`)

- Filters subjects based on matching list
 - Renames columns to match IDOR format
-

2. Data Cleaning Missing values (`replace_nans`)

- Detects and replaces invalid (empty) responses
- Warning : Current behavior:
 - Missing/invalid values are replaced with 1 (default)
 - Median-based imputation is **not yet implemented** (missing information from the clinical psychology team)

3. Score Computation

- Core function: `calculate_SRS2_scores`
- For each subject:

1. Loads `SRS2_correction.xlsx`

2. Uses the following fields to fill the formula :

- test type
- sex
- responses

3. Extracts computed scores

Output scores:

- TOTAL
 - PERCEPÇÃO SOCIAL
 - COGNIÇÃO SOCIAL
 - COMUNICAÇÃO SOCIAL
 - MOTIVAÇÃO SOCIAL
 - PADRÕES RESTRITOS E REPETITIVOS
 - COMUNICAÇÃO E INTERAÇÃO SOCIAL
-

4. Output Files Generated in csv/:

SRS2_results.csv	# Wide format (one row per subject)
SRS2_results_long.csv	# Long format (one row per score, useful for plotting)
SRS2_results_HCP.csv	# Controls

5. Analysis

- file `first_analysis_SRS2`

Performs basic descriptive analyses (to be continued):

- Number of subjects per group
- Mean scores:
 - global
 - per subscale
- Comparison by neuroclassification
- Proportion of subjects above threshold (score > 50)

II - Choosing the control participants

Overview

This script builds a **matched dataset between patients and control subjects** for analyses (behavioral and neuroimaging).

The goal is to:

- Select **patients with valid data** (fMRI + SRS-2 + neuroclassification)
- Select **eligible controls** from HCP datasets
- Match each patient with a control subject based on:
 - **Sex**
 - **Age (as close as possible)**

Usage

Run the script:

```
bash id="t8o4d1" python matching.py
```

Output

Saved in:

```
bash id="y9b7j3" csv/matching.csv
```

Patients (IDOR dataset)

- fMRI data (age at acquisition, gender)
- Neuroclassification
- SRS-2 availability

Controls (HCP dataset)

- Demographic files from HCP
- SRS-2 availability

Pipeline Steps

1. Extract Patient List

`extract_patient_list()` Selects patients that meet all required criteria:

- Have **fMRI data**
- Have a valid **neuroclassification**:
 - Total
 - Partial
 - Hipoplásico
- Have **SRS-2 data available**
- Output: list of eligible patients with:
 - `subj_id`
 - age at acquisition
 - gender

2. Extract Control List

`extract_control_list()` Processes HCP datasets to identify valid control subjects:

- Loads multiple demographic files from HCP
- Keeps only participants with:
 - valid demographic data
 - available **SRS-2**
- Output: list of potential controls with:
 - subject ID
 - age
 - sex

3. Matching Procedure

`compute_matching(fmri, hcp)` Matches each patient to a control subject based on:

- **Sex matching** (strict)
- **Age proximity** (best match selected)

For each patient:

- All possible controls with same sex are considered
- The control with the **closest age** is selected
- Output includes:
 - Patient ID
 - Matched control ID
 - Patient age
 - Control age
 - Age difference

III - SRS-2 Manual Entry & Editing Tool

This script (`enter_SRS.py`) provides a simple **interactive tool to manually enter or correct SRS-2 questionnaire responses**.

What it does

- Lets you **input SRS-2 answers** for a participant (question by question)
- Supports different test versions (preescolar, escolar, adulto)
- Saves responses into a CSV file:
- can edit/verify the responses of one participant

`csv/SRS2_manual.csv`

Main Functions

- `extract_lists` — extracts and records the list of responses for a participant
- `edit_participant` — allows reviewing and modifying existing responses

Purpose

This tool is useful when:

- Data is missing from REDCap
- Manual correction is required